

Technology Stack

Date	30 September 2026
Team ID	SWTID-2026-8550
Project Title	AI BlogNest API

Technology Stack

Technology	Purpose
Node.js	Backend runtime.
Express.js	REST API framework.
MongoDB	Primary NoSQL database.
Mongoose	ODM, schemas and validation.
Google Gemini AI	AI generation and summarization.
JWT	Stateless authentication.
bcrypt.js	Password hashing.
Postman	API verification.
ThunderClient	API testing.
Visual Studio Code	Development environment.
npm	Dependency management.

Technology Architecture

API Client → Node.js → Express.js → Services → Mongoose → MongoDB
→ Google Gemini AI

↘ AI Flow

Technology Selection

Node.js + Express.js provide the backend runtime/framework.

MongoDB provides document storage and Mongoose provides schema-based interaction.

Google Gemini AI supports generation and summarization.

JWT supports authentication and RBAC protects routes.

bcrypt.js protects passwords through hashing.

Postman/ThunderClient support API verification.

Development Environment

Category	Requirement
OS	Windows 10/11, macOS or Linux
Runtime	Node.js v16+
Package Manager	npm v8+
Database	MongoDB
Editor	Visual Studio Code
API Testing	Postman / ThunderClient
Hardware	Intel i5 8th Gen+ / Ryzen 5; 8GB RAM minimum